

What causes it?

Last revised in March 2025

- Type 2 diabetes is caused by a combination of insulin resistance/insensitivity (where the body is unable to respond to normal levels of insulin) and insulin deficiency (where the pancreas is unable to secrete enough insulin to compensate for this resistance) [Mayer-Davis, 2018 (</topics/diabetes-type-2/references/>)] [Zeitler, 2018 (</topics/diabetes-type-2/references/>)].
 - People with type 2 diabetes can differ in the balance of importance of increased insulin resistance and decreased insulin secretion that contributes to their disease, and both mechanisms may co-exist.
 - Insulin resistance may be exacerbated by obesity, physical inactivity, and other [risk factors](/topics/diabetes-type-2/background-information/risk-factors/) (</topics/diabetes-type-2/background-information/risk-factors/>).

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What are the causes and risk factors?

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- **Type 1 diabetes is caused by an absolute insulin deficiency, usually resulting from autoimmune destruction of the insulin-producing beta cells in the pancreas.**
- **Risk factors include:**
 - **Genetic factors** — type 1 diabetes is a heritable polygenic disease.
 - A sibling of a person with type 1 diabetes has 6–7% risk of developing it themselves. This risk increases to 30–70% in identical twins.
 - The risk for a child is 1–9% if a parent has type 1 diabetes.
 - Several different alleles (alternative forms of a gene) and haplotypes (combinations of alleles) have been associated with either an increased or decreased risk of type 1 diabetes. However, the mechanisms by which these haplotypes interact and alter risk are not completely understood.
 - **Environmental factors** — there is evidence that in genetically susceptible people, certain environmental factors can trigger the development of autoimmunity to the pancreatic beta cells.
 - Environmental factors associated with type 1 diabetes include diet, vitamin D exposure, obesity, early-life exposure to viruses associated with islet inflammation (such as enteroviruses), and decreased gut-microbiome diversity.
 - The importance of environmental risk factors (regardless of genetic background risk) are highlighted by the large differences in the incidence of type 1 diabetes in genetically similar populations that are separated by socioeconomic borders and by the increasing incidence of type 1 diabetes in genetically low-risk people.

[[NICE, 2015a \(/topics/diabetes-type-1/references/\)](#); [DiMeglio 2018 \(/topics/diabetes-type-1/references/\)](#); [WHO, 2019 \(/topics/diabetes-type-1/references/\)](#); [Norris, 2020 \(/topics/diabetes-type-1/references/\)](#)]

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What are the complications?

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- **Microvascular complications** (due to damage to small blood vessels)
 - Nephropathy — diabetic kidney disease is caused by damage to small blood vessels in the kidneys.
 - Almost 20% of people with diabetes will need treatment for kidney disease during their lifetime [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
 - At least 10,350 people in the UK have end-stage kidney failure due to diabetes.
 - People with diabetes are 5 times more likely to need either kidney dialysis or a kidney transplant than people without diabetes.
 - Retinopathy — diabetic retinopathy is caused by small blood vessel damage to the retina, leading to progressive loss of vision and possible blindness [WHO, 2020 ([/topics/diabetes-type-1/references/](#))].
 - Diabetes is responsible for 5% of all sight loss in the UK [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
 - Almost 50% of people with type 1 diabetes have some form of diabetic retinopathy; after 20 years of living with the disease, almost everyone with type 1 will have some degree of retinopathy.
 - About 14% of people of working age who have severe visual impairment have diabetes.
 - People with diabetes are more likely to get glaucoma and cataracts, both of which can lead to blindness.
 - Neuropathy — diabetes causes nerve damage through different mechanisms, including direct damage by the hyperglycemia and decreased blood flow to nerves by damaging small blood vessels [WHO, 2020 ([/topics/diabetes-type-1/references/](#))].
 - Painful diabetic neuropathy is estimated to affect between 16–26% of people with diabetes [NICE, 2019c ([/topics/diabetes-type-1/references/](#))].
 - Diabetic foot disease often leads to ulceration and subsequent limb amputation [WHO, 2020 ([/topics/diabetes-type-1/references/](#))]. Between 70,000–90,000 people with diabetes have a foot ulcer in any given week, and diabetes leads to more than 8500 leg, toe, or foot amputations every year [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
 - Autonomic neuropathy is a late-stage complication of diabetes where the neurons of the autonomic nervous system become damaged owing to chronic hyperglycaemia. It presents in different ways and affects a variety of organs, including the skin (sweating), blood vessels (postural hypotension), gastrointestinal tract (gastroparesis and diarrhoea), heart, bladder function, and sexual function. It may also blunt the symptoms of [hypoglycaemia](#) ([/topics/diabetes-type-1/diagnosis/when-to-suspect-hypoglycaemia/](#)). [NICE, 2015a ([/topics/diabetes-type-1/references/](#))].
 - Men with diabetes are 3 times more likely to have trouble getting or keeping an erection than men without diabetes. More than 25% of women with type 1 diabetes have sexual problems [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
- **Macrovascular complications** (due to damage to larger blood vessels)
 - Diabetes is a risk factor for the development of atherosclerosis, which increases the risk of cardiovascular diseases (CVDs), such as myocardial infarction, heart failure,

stroke, and peripheral arterial disease (for example intermittent claudication) [[WHO, 2020 \(/topics/diabetes-type-1/references/\)](#)].

- Type 1 diabetes is associated with a 2–4 fold increased risk of CVD in adults aged over 40 years, especially in the presence of renal disease [[ABCD-RA, 2017 \(/topics/diabetes-type-1/references/\)](#)].
- Compared with people without diabetes, people with type 1 diabetes are more than 4 times as likely to have a heart attack, 4.5 times more likely to experience heart failure, and 3.5 times more likely to have a stroke [[Diabetes UK, 2019 \(/topics/diabetes-type-1/references/\)](#)].

- **Metabolic complications**

- Diabetic ketoacidosis (DKA) is a serious complication of diabetes characterized by hyperglycaemia (or a diagnosis of diabetes), metabolic acidosis, and ketonaemia [[Joint British Diabetes Societies Inpatient Care Group, 2013 \(/topics/diabetes-type-1/references/\)](#); [Karslioglu French, 2019 \(/topics/diabetes-type-1/references/\)](#)].
 - Insulin deficiency results in an increase in counter-regulatory hormone production (glucagon, cortisol, growth hormone, and catecholamines). This type of hormonal imbalance enhances hepatic glycogenolysis (glucose production through the breakdown of glycogen stores) and gluconeogenesis (formation of glucose from other substrates), resulting in severe hyperglycaemia [[Joint British Diabetes Societies Inpatient Care Group, 2013 \(/topics/diabetes-type-1/references/\)](#)].
 - Enhanced lipolysis (due to the lack of utility of glucose) increases serum free fatty acids that are then metabolized as an alternative energy source in the process of ketogenesis, resulting in accumulation of large quantities of ketone bodies and subsequent metabolic acidosis [[Joint British Diabetes Societies Inpatient Care Group, 2013 \(/topics/diabetes-type-1/references/\)](#)].
 - DKA is a medical emergency because it leads to dehydration and electrolyte imbalances [[Karslioglu French, 2019 \(/topics/diabetes-type-1/references/\)](#)]. The mechanisms responsible for fluid depletion in DKA include osmotic diuresis due to hyperglycaemia, vomiting, and inability to take in fluid owing to a diminished level of consciousness. Electrolyte shifts and depletion are in part related to the osmotic diuresis.
- Hypoglycaemia is an inevitable adverse effect of insulin treatment. It is generally defined as being present when blood glucose levels fall to less than 3.5 mmol/L; however, the absolute blood glucose level at which signs and symptoms begin to occur can vary.
 - Severe hypoglycaemia may result in convulsions, inability to swallow, loss of consciousness, and coma. Between 4–10% of deaths in people with type 1 diabetes are attributed to hypoglycaemia [[NICE, 2015a \(/topics/diabetes-type-1/references/\)](#)].
 - See the section on [Hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia\)](#) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](#) for more information on hypoglycaemia.

- **Other autoimmune conditions**

- People with type 1 diabetes are at increased risk of developing other autoimmune diseases, most commonly thyroid disease (Graves' disease or Hashimoto's thyroiditis, 15–30%), autoimmune gastritis and/or pernicious anaemia (5–10%), coeliac disease (4–9%), vitiligo (2–10%), and Addison's disease (0.5%) [[Van den Driessche, 2009 \(/topics/diabetes-type-1/references/\)](#)].

- **Psychological complications**

- Around 40% of people with diabetes experience emotional or psychological problems, such as anxiety, depression, and diabetes-related emotional distress (which affects about 42% of adults with type 1 diabetes). Poor emotional and psychological wellbeing is associated with poorer quality of life, greater difficulty with self management, and sub-optimal glycaemic control [Diabetes UK, 2018 ([/topics/diabetes-type-1/references/](#))].
- In children and young people, psychological complications may include behavioural and conduct disorders, family/relationship difficulties, and risk-taking behaviour (including non-adherence to recommended treatment) [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
- Needle phobias, fear of self-injecting, and eating disorders (which affect over 30% of women in their early twenties with type 1 diabetes) may lead to poor glycaemic control and subsequent complications [Diabetes UK, 2018 ([/topics/diabetes-type-1/references/](#))].
- A position statement (<http://diabetes-resources-production.s3-eu-west-1.amazonaws.com/diabetes-storage/2017-08/Diabulimia%20Position%20statement%20Mar%202017.doc>), issued by Diabetes UK states that around 40% of women and 11% of adolescent boys with diabetes may omit or reduce insulin to lose weight (diabulimia). Diabulimia is not a recognized medical or psychiatric term, but the Diagnostic Statistical Manual of Mental Disorders (DSM-5) considers that insulin omission in order to lose weight is a clinical feature of anorexia and bulimia.
- **Infections and other skin complications**
 - People with diabetes are prone to infections, particularly of the urinary tract and skin [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))], and severe infections can lead to amputations [WHO, 2020 ([/topics/diabetes-type-1/references/](#))].
 - Necrobiosis lipoidica is an uncommon inflammatory condition in which shiny, reddish-brown or yellowish patches develop in the skin of people with diabetes [BAD, 2019 ([/topics/diabetes-type-1/references/](#))].
- **Reduced quality of life**
 - People with diabetes face significant challenges to daily living, such as managing episodes of hypoglycaemia and hyperglycaemia, daily administration of insulin, regular self-monitoring of blood glucose, and the need to plan normal daily activities (such as eating and exercising) [NICE, 2015a ([/topics/diabetes-type-1/references/](#))].
- **Reduced life expectancy**
 - Having type 1 diabetes typically reduces life expectancy in the UK by 11–15 years [NICE, 2015a ([/topics/diabetes-type-1/references/](#)); NICE, 2019b ([/topics/diabetes-type-1/references/](#))].
 - The risk of death is 135% higher than for people of the same age who do not have diabetes. Most of the deaths are due to chronic complications of diabetes, although death from acute hypoglycaemia or DKA may occur [NICE, 2015a ([/topics/diabetes-type-1/references/](#))].

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What are the complications?

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Possible complications of type 2 diabetes include:

- **Macrovascular**

- Atherosclerotic cardiovascular disease (CVD)
 - CVD including stroke and peripheral arterial disease is the leading cause of death in people with type 2 diabetes [[Davies, 2018 \(/topics/diabetes-type-2/references/\)](#)].
 - People with type 2 diabetes have a two-fold increased risk of stroke within the first 5 years of diagnosis compared with the general population [[Diabetes UK, 2019a \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topic on [Stroke and TIA \(/topics/stroke-tia/\)](#) for more information.
 - People with type 2 diabetes are 2.5 times more likely to have a myocardial infarction and develop heart failure than people without diabetes [[Diabetes UK, 2019c \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topics on [Angina \(/topics/angina/\)](#) and [Heart failure - chronic \(/topics/heart-failure-chronic/\)](#) for more information.
 - Peripheral arterial disease is a risk factor for the development of diabetic foot disease and foot ulcers and is present in up to 50% of people with diabetic foot ulcer [[Schaper, 2020 \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topic on [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](#) for more information.

- **Microvascular**

- Diabetic kidney disease
 - This may be caused by diabetic nephropathy (hyperglycaemia-induced glomerular disease resulting in albuminuria in the absence of other causes of albuminuria), hypertension, and renal atheroma or ischaemia, which cause varying degrees of glomerulosclerosis and tubular fibrosis [[Mark, 2017 \(/topics/diabetes-type-2/references/\)](#); [Lewis, 2019 \(/topics/diabetes-type-2/references/\)](#)]. Proteinuria is usually the first sign of diabetic kidney disease and is a marker of increased cardiovascular risk [[Lewis, 2019 \(/topics/diabetes-type-2/references/\)](#)].
 - Diabetes is the commonest cause of chronic kidney disease (CKD) and end-stage renal disease (ESRD), which may require renal replacement therapy or kidney transplantation [[Winocour, 2020 \(/topics/diabetes-type-2/references/\)](#)]. Diabetes is the most commonly identified primary renal disease for people starting renal replacement therapy [[Renal Association, 2020 \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](#) for more information.
- Retinopathy
 - Diabetes is the leading cause of preventable sight loss in the UK, and 7% of people who are newly registered blind in England and Wales have lost their sight due to diabetes [[Diabetes UK, 2019c \(/topics/diabetes-type-2/references/\)](#)].
 - More than one-quarter of people with type 2 diabetes have some form of retinopathy [[Diabetes UK, 2019a \(/topics/diabetes-type-2/references/\)](#)].
 - Within 20 years of diagnosis, two-thirds of people with type 2 diabetes will have some degree of retinopathy [[Diabetes UK, 2019a \(/topics/diabetes-type-](#)

[2/references/](#)).

- Peripheral neuropathy
 - This is estimated to affect up to 50% of people with diabetes, and commonly presents as distal and symmetrical polyneuropathy [[Guttormsen, 2017 \(/topics/diabetes-type-2/references/\)](#)].
 - Painful neuropathy may cause symptoms of numbness, burning or shooting pain, tingling and/or paraesthesia of the hands and/or feet typically in a stocking and glove distribution, often at night. Over time it can progress to persistent neuropathic pain [[Donoghue, 2018 \(/topics/diabetes-type-2/references/\)](#); [Baker, 2020 \(/topics/diabetes-type-2/references/\)](#)].
- Autonomic neuropathy
 - This can affect multiple systems and includes postural hypotension, gastroparesis (delayed gastric emptying), unexplained diarrhoea, inadequate bladder emptying, sexual dysfunction including erectile dysfunction, sweating abnormalities, and impaired awareness of hypoglycaemia [[Donoghue, 2018 \(/topics/diabetes-type-2/references/\)](#); [NICE, 2020a \(/topics/diabetes-type-2/references/\)](#)].
- **Foot problems**
 - Diabetic foot problems may result from a combination of peripheral arterial disease and peripheral neuropathy, due to loss of protective foot sensation and possible deformity leading to abnormal loading of the foot. Minor trauma, for example from ill-fitting shoes or injury can precipitate foot ulceration [[Baker, 2020 \(/topics/diabetes-type-2/references/\)](#); [Schaper, 2020 \(/topics/diabetes-type-2/references/\)](#)].
 - Limb- or life-threatening complications of foot disease include foot ulceration with signs of sepsis, lower limb ischaemia risking limb amputation, deep-seated soft tissue or bone infection (osteomyelitis), and gangrene [[NICE, 2019b \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topics on [Sepsis \(/topics/sepsis/\)](#) and [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](#) for more information.
- **Metabolic**
 - Dyslipidaemia
 - Type 2 diabetes is commonly associated with dyslipidaemia, which is a risk factor for CVD [[NICE, 2020a \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topics on [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](#) and [CVD risk assessment and management \(/topics/cvd-risk-assessment-management/\)](#) for more information.
 - Diabetic ketoacidosis (DKA)
 - This is the metabolic state characterized by the triad of hyperglycaemia, metabolic acidosis, and ketonaemia. It is a life-threatening emergency that may lead to dehydration and electrolyte imbalances, myocardial infarction, shock, cerebral oedema, and aspiration pneumonia. Although more common in type 1 diabetes, people with type 2 diabetes can also develop DKA [[French, 2019 \(/topics/diabetes-type-2/references/\)](#); [BSPED, 2020 \(/topics/diabetes-type-2/references/\)](#)].
 - People with a history of recurrent DKA are at high risk for all-cause mortality [[French, 2019 \(/topics/diabetes-type-2/references/\)](#)].
 - Hyperosmolar hyperglycaemic state (HHS)
 - This is a rare but potentially life-threatening emergency that occurs more typically in older people with type 2 diabetes and other co-morbidities, who develop severe hyperglycaemia, often over several weeks, often due to a combination of acute illness and dehydration. It may present with urinary frequency, thirst, nausea, confusion, and severe dehydration [[Down, 2020 \(/topics/diabetes-type-2/references/\)](#)].

- **Psychosocial**
 - Diabetes may negatively impact on quality of life and is associated with an increased risk of anxiety and depression. The prevalence of depression is about twice as high in people with diabetes as in the general population [Milne, 2020 ([/topics/diabetes-type-2/references/](#))]. It may be exacerbated by other factors such as chronic painful peripheral neuropathy [Guttormsen, 2017 ([/topics/diabetes-type-2/references/](#))]. See the CKS topics on [Generalized anxiety disorder \(/topics/generalized-anxiety-disorder/\)](#) and [Depression \(/topics/depression/\)](#) for more information.
 - Children and young people may experience depression, behavioural problems, family/relationship difficulties, and increased risk-taking behaviour (including non-adherence to treatment) [Candler, 2018a ([/topics/diabetes-type-2/references/](#)); Zeitler, 2018 ([/topics/diabetes-type-2/references/](#))]. See the CKS topics on [Depression in children \(/topics/depression-in-children/\)](#) and [Conduct disorders in children and young people \(/topics/conduct-disorders-in-children-young-people/\)](#) for more information.
- **Reduced life expectancy**
 - It is estimated that diabetes accounts for about 15–16% of deaths in England, with life expectancy for people with type 2 diabetes reduced by an average of up to 10 years. Diabetes UK estimates that in people with type 2 diabetes, the risk of death is 32% greater than expected, and is mostly in younger people and females [Diabetes UK, 2015 ([/topics/diabetes-type-2/references/](#))].

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What is it?

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- **Diabetes mellitus** is a metabolic disorder characterized by persistent hyperglycaemia (random plasma glucose more than 11 mmol/L) with disturbances of carbohydrate, protein, and fat metabolism resulting from defects in insulin secretion (leading to insulin deficiency), insulin action (leading to insulin resistance), or both.
- **In type 1 diabetes**, an absolute insulin deficiency causes persistent hyperglycaemia.
 - The term 'type 1 diabetes' has replaced the older terms 'insulin-dependent diabetes mellitus (IDDM)' and 'juvenile-onset diabetes' because the older terms are potentially misleading: type 1 diabetes can also develop during adulthood and some people with type 2 diabetes are treated with insulin.
- **Other types of diabetes include:**
 - **Type 2 diabetes** — insulin resistance and a relative insulin deficiency result in persistent hyperglycaemia. See the CKS topic on [Diabetes – type 2 \(/topics/diabetes-type-2/\)](#) for more information.
 - **Gestational diabetes** — hyperglycaemia develops during pregnancy and usually resolves after delivery, although the woman is at increased risk for overt type 2 diabetes in the future.
 - **Other specific types of diabetes, including:**
 - Monogenic diabetes (due to a single gene defect; previously known as 'maturity-onset diabetes in the young').
 - Diabetes secondary to pathological conditions or diseases (such as pancreatitis, trauma, or pancreatic surgery).
 - Drug- or chemically-induced diabetes (such as from long-term corticosteroids or antipsychotics).

[NICE, 2016a ([/topics/diabetes-type-1/references/](#)); SIGN, 2017 ([/topics/diabetes-type-1/references/](#)); WHO, 2019 ([/topics/diabetes-type-1/references/](#))]

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What is it?

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- Diabetes mellitus is a group of metabolic disorders in which persistent hyperglycaemia (random plasma glucose more than 11.1 mmol/L) is caused by deficient insulin secretion, resistance to the action of insulin, or both [[Mayer-Davis, 2018 \(/topics/diabetes-type-2/references/\)](#)] [[WHO, 2019 \(/topics/diabetes-type-2/references/\)](#)].
 - Insulin deficiency and insulin resistance lead to the abnormalities of carbohydrate, fat, and protein metabolism that are characteristic of diabetes mellitus [[Mayer-Davis, 2018 \(/topics/diabetes-type-2/references/\)](#)].
- Diabetes mellitus is classified as [[Mayer-Davis, 2018 \(/topics/diabetes-type-2/references/\)](#)] [[Zeitler, 2018 \(/topics/diabetes-type-2/references/\)](#)] [[WHO, 2019 \(/topics/diabetes-type-2/references/\)](#)] [[NICE, 2020a \(/topics/diabetes-type-2/references/\)](#)]:
 - Type 1 diabetes — an absolute insulin deficiency causes persistent hyperglycaemia (insulin activity is normal). See the CKS topic on [Diabetes – type 1 \(/topics/diabetes-type-1/\)](#) for more information.
 - Type 2 diabetes — insulin resistance and a relative insulin deficiency result in persistent hyperglycaemia.
 - Gestational diabetes — hyperglycaemia develops during pregnancy and usually resolves after delivery, although the woman is at increased risk for overt type 2 diabetes in the future.
 - Other specific types of diabetes, including:
 - Monogenic diabetes (caused by specific gene mutations, some affecting pancreatic beta-cell function, others affecting insulin action). The previously known 'maturity-onset diabetes of the young' presents with an autosomal dominant family history of diabetes, with typically mild, non-ketotic disease presenting in adolescence or early adulthood.
 - Secondary to conditions or diseases of the exocrine pancreas, such as pancreatitis, malignancy, trauma or surgery, cystic fibrosis, or haemochromatosis, for example.
 - Endocrine disorders, such as acromegaly, Cushing's syndrome, hyperthyroidism, or pheochromocytoma.
 - Drug- or chemical-induced, such as following corticosteroid treatment.
- The term 'type 2 diabetes' has replaced the terms 'non-insulin-dependent diabetes mellitus (NIDDM)' and 'adult-onset diabetes', as people with type 2 diabetes may require insulin, and type 2 diabetes is increasingly being diagnosed in children and young people.

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